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# The Effect of Folic Acid on cAMP-Elicited cAMP Production in *Dictyostelium discoideum*

PETER N. DEVREOTES

*Department of Physiological Chemistry, Johns Hopkins University School of Medicine, 725 N. Wolfe Street, Baltimore, Maryland 21205*

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In *Dictyostelium discoideum*, folic acid serves as a chemoattractant, accelerates the appearance of several developmentally regulated enzymes, alters the phase of spontaneous light-scattering oscillations, and elicits increases in cGMP and cAMP. The effect of folic acid and pteridine analogs on rates of [$^3$H]cAMP secretion, monitored by rapid perfusion, was studied. At low cell densities folic acid elicited only minimal [$^3$H]cAMP secretion. At high cell densities, folic acid elicited greater increases in cellular cAMP and rates of [$^3$H]cAMP secretion than did maximally effective doses of cAMP. Following pretreatment with $10^{-8}$ *M* cAMP for 10 min or $10^{-6}$ *M* cAMP for 2 min, no [$^3$H]cAMP was secreted in response to folic acid. Conversely, pretreatment with folic acid did not impair subsequent responsiveness to $5 \times 10^{-9}$ *M* cAMP. Simultaneous addition of folic acid and cAMP increased the release of [$^3$H]cAMP two- to threefold compared to control stimuli of the same concentration of cAMP. Pretreatment with a mixed stimulus attenuated the response to a subsequent increment in the cAMP concentration only if the higher cAMP stimulus did not include folic acid. The effects of folic acid are mimicked by the pteridine ring alone; analogs substituted at the 2- or 5-position are ineffective; non-reducible analogs are partially active. The observations presented here suggest significant differences in the mechanisms by which cAMP and folic acid stimulate cAMP production. Rather than there being separate receptors for each chemoattractant that interact in the same way with an intracellular pathway, it appears that folic acid and pteridine analogs potentiate the response to cAMP.

## INTRODUCTION

*Dictyostelium discoideum* grows as free-living amoebae, until the nutrient supply is exhausted or removed. Starvation triggers a developmental program; initially the amoebae become specialized to communicate via extracellular 3′,5′-cyclic adenosine monophosphate (cAMP). Chemotaxis toward the nucleotide mediates the aggregation of up to $10^6$ cells to form multicellular structures. Time-lapse observation of this highly coordinated process suggests that the chemotactic signals are relayed through the monolayer for long distances (up to 3 cm). Previous work has shown that isolated amoebae are attracted to cAMP and secrete cAMP in response to exogenous cAMP stimuli (Konijn *et al.*, 1968; Shaffer, 1975; Roos *et al.*, 1975). Furthermore, waves of cAMP, which emanate from centers with a regular periodicity of 5 to 10 min have been visualized in monolayers of aggregating amoebae (Tomchik and Devreotes, 1981; reviewed in Devreotes, 1982).

The cAMP-signaling response (i.e., cAMP-elicited cAMP secretion) has been extensively characterized. Figure 1 illustrates the interaction of the components which generate the cAMP-signaling response. A change in occupancy of surface cAMP receptors leads to activation of adenylate cyclase (V, in Fig. 1) and an ensuing accumulation of intracellular cAMP (Roos *et al.*, 1975; Dinauer *et al.*, 1980a; Gerisch and Wick, 1975). Not all of the newly synthesized cAMP can be accounted for in the extracellular medium, suggesting that a fraction is degraded by intracellular phosphodiesterase (Dinauer *et al.*, 1980a). The cAMP secretion rate is directly proportional to the intracellular cAMP level; there is no long-term storage of newly synthesized cAMP (Gerisch and Wick, 1975; Dinauer *et al.*, 1980a). After several minutes, the rate of cAMP synthesis decreases (see below) and the remaining cAMP is degraded by extracellular and membrane-bound phosphodiesterase.

Interaction of extracellular cAMP with surface cAMP receptors can create a positive feedback loop since the secreted cAMP adds to the stimulus. Extracellular phosphodiesterase counteracts this feedback loop by decreasing extracellular cAMP; removal (or decrease) of the cAMP stimulus during a cAMP-signaling response leads to the abrupt termination of the response (Devreotes and Steck, 1979). The opposing effects of the positive feedback loop and phosphodiesterase suggest the system can exhibit excitable behavior. Evidence for this excitability is found when the cells are studied in suspension (and presumably occurs *in situ* on an agar surface). For instance, at critical times in development cAMP levels oscillate spontaneously (Gerisch and Hess, 1974; Gerisch and Wick, 1975). Furthermore, there is a discrete "refractory period" before a second reponse can be initiated.

The effects of both the positive feedback loop and the extracellular phosphodiesterase can be minimized if the cells are stimulated under conditions of rapid perfusion.



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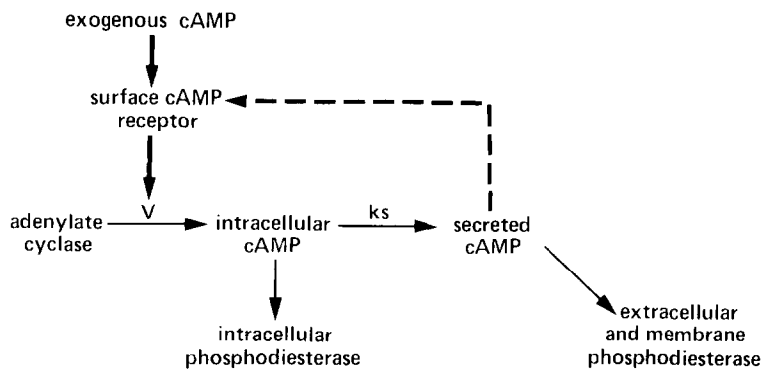


FIG. 1. Interaction of components involved in generation of cAMP signaling response. Activity of adenylate cyclase, V, is modulated by occupancy of surface receptors. Rate of cAMP secretion is directly proportional to intracellular concentration ($k_S$ is first-order rate constant). Dotted line represents autocatalytic feedback loop whereby secreted cAMP can serve as an additional stimulus. Exogenously added cAMP is also subject to degradation by extracellular and membrane phosphodiesterases.

Secreted cAMP is rapidly removed; degraded cAMP is rapidly renewed. In this way, extracellular cAMP is "clamped" at the concentration of the applied stimulus (Devreotes *et al.*, 1979). However, even when a cAMP stimulus is held constant, the activation of adenylate cyclase is transient. The intracellular cAMP level and the rate of cAMP secretion peak after 2 min of stimulation and then fall to basal values after 5–10 min. The mechanism by which amoebae terminate their signaling response to extracellular cAMP is referred to as adaptation. This mechanism presumably serves to prevent cells from responding indefinitely to their own secretions, which would disrupt intercellular signaling. Adaptation appears to reflect the adjustment of cell sensitivity to the ambient concentration of extracellular cAMP in that adapted cells become unresponsive to a given cAMP stimulus, yet respond to further increases in the external cAMP concentration. Adaptation persists as long as extracellular cAMP is present, but can be reversed by removing the stimulus.

Folic acid (pteroylglutamic acid) has been implicated as a chemoattractant in the early developmental stages of *D. discoideum* and other cellular slime molds (Pan *et al.*, 1972, 1975). As the cells differentiate, folic acid deaminase appears in parallel with the appearance of extracellular cAMP phosphodiesterase (Pan and Wurster, 1978; Bernstein and Van Driel, 1980a,b). Folic acid and cAMP each have been shown to induce the precocious appearance of these enzymes (Klein, 1975; Tsang and Coukell, 1977; Bernstein and Van Driel, 1980a,b; Bernstein *et al.*, 1981). Addition of nanomolar to micromolar amounts of folic acid (or cAMP) at 5- to 7-min intervals accelerates the appearance of spontaneous oscillations in cellular cAMP levels. The phase of these spontaneous oscillations is altered in a similar manner by each of the chemoattractants (Wurster and Schubiger, 1977; Wurster *et al.*, 1977; Gerisch *et al.*, 1977). Folic acid also elicits increases in cellular cGMP levels and light-scattering responses (i.e., stimulus-induced decreases in optical density) in suspensions of *D. discoideum* and related species (Mato *et al.*, 1977; Wurster and Schubiger, 1977; Wurster *et al.*, 1978). In addition it was shown that folic acid can elicit large changes in cellular cAMP levels (Wurster *et al.*, 1978). The latter finding led to the experiments reported here.

Based on the apparent similarities of many of the effects of folic acid and cAMP, it has been proposed that cell surface receptors for folic acid interact with an intracellular pathway similar to that linked to cell surface cAMP receptors (Bernstein *et al.*, 1981). Indeed, the association of [$^3$H]folic acid and [$^3$H]methotrexate with *D. discoideum* cells has been measured (Wurster and Butz, 1980; Van Driel, 1981 Nandini-Kishore and Frazier, 1982). In an effort to determine the extent of overlap of the pathways for activation of adenylate cyclase by folic acid and cAMP, cAMP synthesis and secretion elicited by each of these chemoattractants were compared. It is clear from the results presented below that the mechanisms by which folic acid and cAMP elicit the response differ dramatically. Folic acid provides only a minimal (perhaps absent) stimulus and even the highest concentrations elicit little or no adaptation. Rather than directly activating the adenylate cyclase, it appears that folic acid potentiates the stimulatory effect of cAMP.

## METHODS

### *Cell Culture and Development*

*D. discoideum* strain NC-4 was grown on SM nutrient agar [10 g bactopeptone, 10 g dextrose, 1 g yeast extract, per liter development buffer (5 m*M* $Na_2HPO_4$, 5 m*M* $KH_2PO_4$, 1 m*M* $MgCl_2$, 200 $\mu M$ $CaCl_2$)] to a density of

1-5 × $10^8$ cells/100-mm dish (Sussman, and Sussman, 1967). No signs of clearing of the bacterial lawn were apparent when the cells were harvested. Cells were freed of bacteria by three to four washes (600*g* for 3 min) in development buffer. Strain Ax-3 was grown in HL-5 medium, harvested at a density less than 5 × $10^6$ cells/ml, and washed twice in development buffer. Cells were shaken at 150 rpm at 22°C for 3-5 hr.

### *cAMP Levels in Cell Suspensions*

Ax-3 cells were washed once, suspended at a density of $10^8$ cells/ml in development buffer, and synchronized by addition of $2 \times 10^{-8}$ *M* cAMP (1/100 volume of 2 × $10^{-6}$ *M*) at 6-min intervals. Cellular cAMP levels were monitored during the fifth and sixth cycles. Fifteen microliters of cell suspension was added to 15 μl of 2 × $10^{-7}$ *M* [$^3$H]cAMP (NEN 38 Ci/mmole) in 5% TCA. Samples were cooled to 4°C and 80 μl of erythrocyte ghost suspension in 0.5 *M* potassium phosphate, pH 7.0, was added (Hesse *et al.*, 1975). After 2 to 18 hr at 4°C, samples were diluted with 5 ml cold 20 m*M* potassium phosphate, pH 6.2, and centrifuged at 1500*g* for 20 min. The pellet was extracted twice with 0.4 ml 1% SDS, 10 m*M* Tris-HCl, pH 7.5, mixed with 7 ml Liquiscint, and counted. Control values were about 23,000 cpm and addition of 20 pmole of unlabeled cAMP lowered the value to about 3000 cpm.

### *[$^3$H]cAMP Secretion in Rapid Perfusion*

Washed NC-4 cells were prepared for measurements of [$^3$H]cAMP secretion exactly as previously described (Dinauer *et al.*, 1980a). In a few cases, development buffer was substituted for nonnutrient buffer. Ax-3 cells were fed [$^3$H]adenosine-labeled *Escherichia coli* following the same protocol used for NC-4 cells. After 3 to 5 hr in shaken suspension, Ax-3 cells were freed of undigested bacteria and added directly to the perfusion apparatus. In each case, cells were synchronized as previously described (Dinauer *et al.*, 1980a).

### *Phosphodiesterase Activity*

Phosphodiesterase activity was measured by a modification of the method of Brooks *et al.* (1968). Reaction mixtures contained 4 × $10^{-5}$ *M* [$^3$H]cAMP, 50 μg/ml 5′-nucleotidase in 100 μl 50 m*M* Tris-HCl, pH 7.5, 5 m*M* $MgCl_2$. The reaction was started by addition of 100-μl cells or extract. To stop the reaction, 100 μl was added to 100 μl methanol and loaded on BioRad AG-2 columns (1 × 0.5 cm) equilibrated in methanol. Columns were eluted with 600 μl methanol directly into scintillation vials.

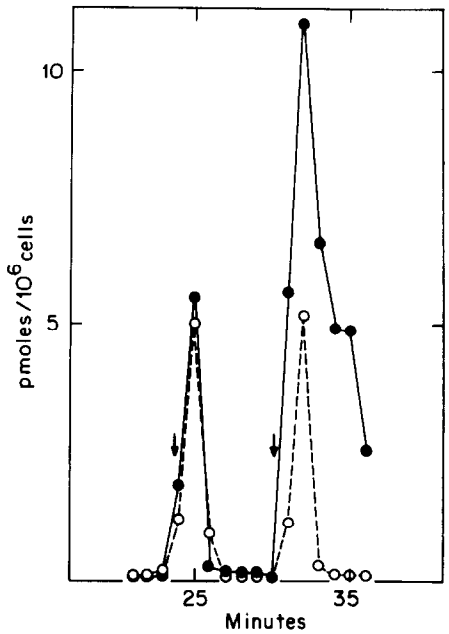


FIG. 2. Elevation in total cAMP levels in suspended Ax-3 cells elicited by cAMP or folic acid stimuli. Preparation of Ax-3 cells and measurement of total cAMP levels are described under Methods. Two parallel cultures were synchronized by addition of 2 × $10^{-8}$ *M* cAMP (final concentration) at 0, 6, 12, 18, 24 min (only the last cycle is shown). At 30 min one culture was adjusted to 2 × $10^{-8}$ *M* cAMP (○); the other to $10^{-5}$ *M* folic acid (●).

## RESULTS

### *Folic Acid-Elicited cAMP Production and Secretion*

It was previously reported that suspensions of aggregation competent *D. discoideum* cells, strain Ax-2, respond to the chemoattractant folic acid by elevating cellular cAMP levels. A similar response was elicited in shaken suspensions of strain Ax-3 following synchronization with nanomolar amounts of cAMP. The responses elicited by cAMP and by folic acid are compared in Fig. 2. Those elicited by folic acid were consistently larger (one- to threefold) and of longer duration (4 to 5 min compared to 2 to 3 min) than those elicited by cAMP. In a few instances, the peak cAMP level occurred 1 min later when folate was used as a stimulus. The response elicited by cAMP plus folic acid was similar to that elicited by folic acid alone.

What is the mechanism by which folic acid increases cAMP levels in cell suspensions? As illustrated in Fig. 1, cAMP synthesis and secretion can be initiated by an abrupt increase in cellular sensitivity or inhibition of phosphodiesterase as well as by the customary stimulus (i.e., an increase in extracellular cAMP). Accordingly, folic acid may initiate a response by inhibition of phosphodiesterase or enhancement of cellular sensitivity rather than by direct activation of adenylate cyclase.

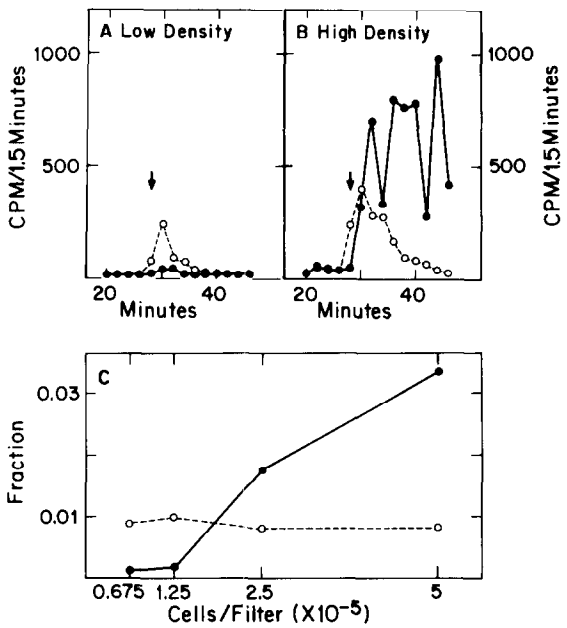


FIG. 3. Effect of cell density on rate of [$^3$H]cAMP secretion from Ax-3 cells elicited by cAMP or folic acid. Cells were prepared and rates of [$^3$H]cAMP secretion were measured as described under Methods. (A, B) Rate of cAMP or folic acid-stimulated [$^3$H]cAMP secretion at low ($1.25 \times 10^5$ cells/filter) or high ($5 \times 10^5$ cells/filter) cell density. Perfusion solution was switched to one containing $10^{-5}$ *M* cAMP (○) or $10^{-4}$ *M* folic acid (●) at the fraction indicated by arrow and then maintained. (C) Amount of [$^3$H]cAMP secreted in 13.5 min following introduction of a $10^{-5}$ *M* cAMP (○) or $10^{-4}$ *M* folic acid (●) stimulus was integrated and expressed as fraction of total radioactivity on filter (i.e., number of cells).

In addition, the positive feedback loop amplifies the initial reaction. Thus, the "response" elicited by folic acid in a cell suspension is necessarily a response to folic acid plus cAMP. To study the direct effect of folic acid on cAMP production, rates of [$^3$H]cAMP secretion from sensitive cells were studied in a perfusion apparatus. The effects of both the positive feedback loop and phosphodiesterase are minimized when cells are rapidly perfused with the cAMP stimulus (Devreotes *et al.*, 1979).

The changes in [$^3$H]cAMP secretion from aggregation competent Ax-3 cells during perfusion with folic acid depended on cell density, perfusion rate, and the particular clone used. As shown in Fig. 3A, at low densities and high perfusion rates the rate of [$^3$H]cAMP secretion increased only slightly when folic acid was added to the perfusion buffer. The increased rate of [$^3$H]cAMP secretion was about 3 to 5% of that elicited by cAMP.

This response differed dramatically from that elicited from Ax-3 cells in suspension (i.e., Fig. 2). However, at high densities and slow rates of perfusion, folic acid elicited changes in [$^3$H]cAMP secretion rates from strain Ax-3 that were reminiscent of those observed in cell suspensions (Figs. 3B, 4E, 5F). Often, the rate of [$^3$H]cAMP secretion increased to a peak value at 3 min which was greater than the peak [$^3$H]cAMP secretion rate elicited by $10^{-5}$ *M* cAMP (Figs. 3B, 4D, E). The rate sometimes decreased to a level about half the peak value and oscillated about a plateau value. There was little tendency of the secretion rate to return toward and remain at the prestimulus value (i.e., no evidence of adaptation). In other Ax-3 clones the folic acid-elicited rate of [$^3$H]cAMP secretion returned nearly to the prestimulus value after 8 to 10 min. As perfusion with folic acid was continued, a second increase in [$^3$H]cAMP secretion often ensued (Devreotes, 1982). Such a secondary increase has never been observed during continued perfusion with cAMP.

Figure 3C summarizes the cell density dependence of folic acid on cAMP-elicited [$^3$H]cAMP secretion from one clone of strain Ax-3. The magnitude of the response per cell was independent of cell density when cAMP was used as a stimulus. In contrast, the amount of [$^3$H]cAMP secreted per cell depended markedly on cell density when folic acid was the stimulus. There was little response elicited by folic acid at low densities ($<1.25 \times 10^5$ cells/cm$^2$) while at slightly higher densities ($\geq 2.5 \times 10^5$ cells/cm$^2$), two to three times more [$^3$H]cAMP was secreted in response to folic acid than in response to cAMP.

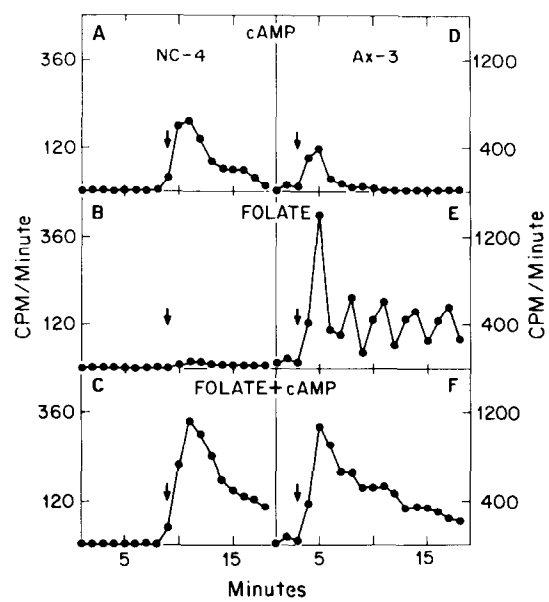


FIG. 4. Comparison of rates of [$^3$H]cAMP secretion elicited by cAMP, folic acid, and cAMP plus folic acid in strains NC-4 and Ax-3. Cells were prepared and rates of [$^3$H]cAMP secretion were measured as described under Methods. Panels A–C were identical sets of NC-4 cells; D–F identical sets of Ax-3 cells. Cell density was $5 \times 10^5$ cells/filter. Stimuli were introduced into the perfusion buffer starting in the fraction indicated by the arrow and then maintained. (A, D) $10^{-5}$ *M* cAMP; (B, E) $10^{-4}$ *M* folic acid; (C, F) $10^{-5}$ *M* cAMP + $10^{-4}$ *M* folic acid.

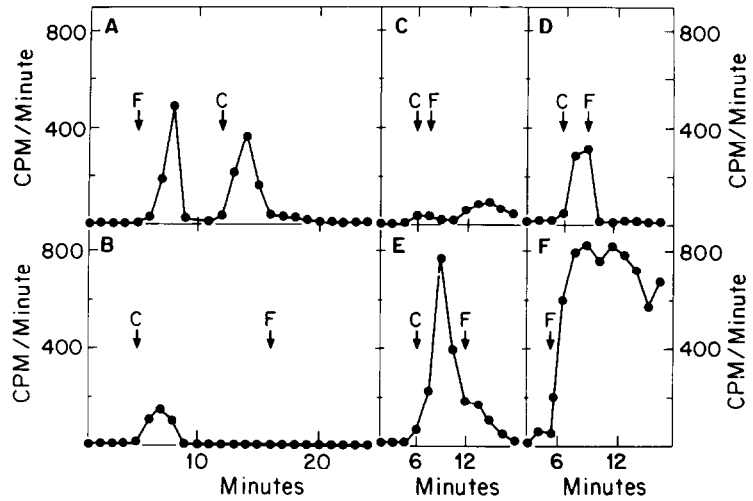


FIG. 5. Effect of pretreatment with cAMP on responsiveness to folic acid in strain Ax-3. Cells were prepared and rates of [$^3$H]cAMP secretion measured as described under Methods. Panels A and B are identical cultures from one experiment; panels C–F identical cultures from an independent experiment. A high cell density was used in each case ($5 \times 10^5$ cells/filter). Perfusion solution was switched to $10^{-4}$ *M* folic acid (F↓) or cAMP (C↓) beginning with fractions indicated. Cyclic AMP concentration was $10^8$ *M* in panel B and $10^{-5}$ *M* in other panels.

The strong dependence on cell density can be interpreted according to the following hypothesis. The response to folic acid is dependent on secreted cAMP. With Ax-3 cells at lower densities (Fig. 3A), secreted cAMP is efficiently removed by rapid perfusion; folic acid alone has only a minimal effect. At higher densities (Figs. 3B, 4E), secreted cAMP quickly accumulates and serves as an additional stimulus which amplifies the initial effect of folic acid into a large "response." This is also consistent with the observations of Ax-3 cells in suspension, where secreted cAMP can accumulate. The effect of folate there is to trigger a large increase in cellular cAMP; folate plus cAMP elicits the same increase (Fig. 2).

NC-4 cells have less a tendency than Ax-3 to spontaneously secrete cAMP and thus provide a cleaner system to study the response to an applied stimulus. Even at higher cell densities, rapid perfusion effectively controls the effects of the positive feedback and phosphodiesterase (Devreotes *et al.*, 1979). Figures 4A and B compare the rates of [$^3$H]cAMP secretion from aggregation competent NC-4 cells elicited by $10^{-5}$ *M* cAMP, a saturating dose, or by $10^{-4}$ *M* folic acid. Under conditions of rapid perfusion, the maximum effect of folic acid is only about 5% of that of cAMP. In seven independent experiments, the rate of [$^3$H]cAMP secretion in response to folic reached a value of 5 to 15% that elicited by $10^{-5}$ *M* cAMP. Note again that when Ax-3 cells at a similar density were perfused with folic acid the response is larger than that elicited by cAMP (Figs. 4D, E; and Fig. 3).

### *Effect of Pretreatment with cAMP on Responsiveness to Folic Acid*

It was next determined whether folic acid would elicit a response in cells that had been rendered less sensitive to cAMP by pretreatment with cAMP (i.e., adapted cells). Figure 5B shows that following pretreatment with $10^{-8}$ *M* cAMP, $10^{-4}$ *M* folic no longer elicits an increase in [$^3$H]cAMP secretion. In addition, 2 min of pretreatment with $10^{-6}$ *M* cAMP was sufficient to extinguish the effects of $10^{-4}$ *M* folic acid (Figs. 5D and E) and a 1-min pretreatment significantly attenuated the effect (Fig. 5C). In contrast, $10^{-4}$ *M* folic acid had little effect on subsequent responsiveness to cAMP (Fig. 5A and see below).

### *Folic Acid Enhances cAMP-Elicited cAMP Secretion*

The data presented thus far indicate that when build up of extracellular cAMP is prevented, folic acid elicits only minimal increases in [$^3$H]cAMP secretion. When cells are in close proximity these small increases can be amplified (i.e., secreted cAMP serves as an additional stimulus) into rates of [$^3$H]cAMP secretion greater than those elicited by a saturating dose of cAMP. This suggests that folic acid increases cellular sensitivity to cAMP or otherwise enhances the response to cAMP. To directly demonstrate this effect, changes in [$^3$H]cAMP secretion rates elicited by folic acid plus cAMP were compared to those elicited by cAMP alone. As shown in Figs. 4C, F and 6A–C, the combined stimulus elicited larger increases in the rate of [$^3$H]cAMP secretion than

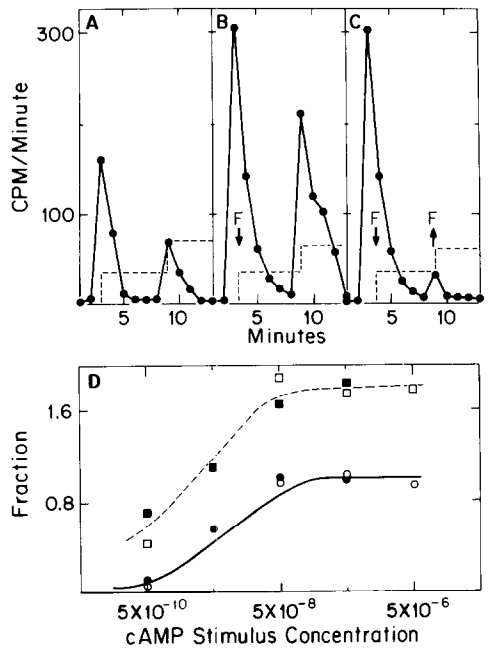


FIG. 6. Enhancement of cAMP-elicited rate of [$^3$H]-cAMP secretion by folic acid. NC-4 cells were prepared and rates of [$^3$H]cAMP secretion measured as described under Methods. (A–C) Three identical cultures were stimulated with $4 \times 10^{-9}$ *M* then $2 \times 10^{-8}$ *M* cAMP as illustrated by the dashed lines. As indicated by the arrows (F↓) folic acid was included with both cAMP stimuli in panel B and only in the $4 \times 10^{-9}$ *M* stimulus in panel C. (D) Total amount of [$^3$H]cAMP secreted as a function of concentration cAMP stimulus in the absence (○, ●) and presence (□, ■) of $10^{-4}$ *M* folic acid. Filled or open symbols represent independent experiments. All data in each experiment were normalized to the $5 \times 10^{-7}$ *M* point.

did cAMP alone. In addition, the increase elicited by the combined stimulus was frequently greater than the algebraic sum of the increases elicited by each chemoattractant individually (Figs. 4A–C). As illustrated in Fig. 6D, this effect was observed with every concentration of cAMP tested. The apparent fold-stimulation by folic acid was greatest at the lowest cAMP stimulus concentrations. In the limit of there being no cAMP stimulus there was still a small increase (i.e., 3 to 5%) elicited by folic acid alone.

### *Effect of Folic Acid on Adaptation*

The potentiation of the cAMP-elicited response by folic acid indicates that the combined stimulus provides a greater positive signal to the adenylate cyclase. It is possible that the combined stimulus also raises cellular adaptation to a higher level. To test this possibility cells were stimulated with serial increments in cAMP concentration in the absence and presence of folic acid. As shown in Figs. 6A and B, when the cAMP stimuli included folic acid, larger increases in [$^3$H]cAMP secretion were observed in response to each increment. However, when the stimulus was switched from $4 \times 10^{-9}$ *M* cAMP plus folic acid to $2 \times 10^{-8}$ *M* cAMP, little increase in the cAMP secretion rate was observed (Fig. 6C). Pretreatment with folic acid plus the lower concentration of cAMP had apparently established a level of adaptation sufficient to significantly reduce an effect of the increment in cAMP stimulus. In another series of experiments it was shown that following complete adaptation to $5 \times 10^{-8}$ *M* cAMP, further secretion of [$^3$H]cAMP could be elicited by changing the stimulus to $5 \times 10^{-8}$ *M* cAMP plus folic acid (not shown).

The above experiments demonstrate that folic acid plus a given concentration of cAMP is interpreted as a somewhat higher cAMP concentration—in terms of both activation of cAMP production and adaptation (Fig. 6). However, previous experiments suggested that folic acid alone elicits little adaptation (Figs. 4E, 5A, F). To test definitively whether folic acid elicits any adaptation, cells were pretreated with $10^{-4}$ *M* folic acid and then tested with a low concentration of cAMP. It appears that even under rapid perfusion conditions, the low level of [$^3$H]cAMP secreted induces some adaptation. This might explain why [$^3$H]cAMP secretion responses during folic acid applications are initially self-limiting but often spontaneously reoccur as the treatment is continued (Fig. 4E). This complication was minimized by two protocols used to test whether folic acid caused adaptation. In the first, perfusion with folic acid was continued well beyond the initial release of [$^3$H]cAMP before the test cAMP stimulus was applied. In the second, the initial release of [$^3$H]cAMP was blocked by caffeine (which is known not to block adaptation—Burton-Theibert and Devreotes, manuscript in preparation). As illustrated in Fig. 7, folic acid pretreatment did not attenuate the subsequent response to cAMP in either protocol. It clearly did not reduce responsiveness as would pretreatment with cAMP.

### *Effect of Folic Acid and Pteridine Analogs on cAMP Secretion*

Increases in the rate of secretion of [$^3$H]cAMP could also be elicited by pteridine and folic acid analogs. Table 1 summarizes these data. The effect of folic acid was dose dependent; with aggregation competent cells, it was first observed with $10^{-8}$ to $10^{-7}$ *M*. The maximum effect occurred at $10^{-4}$ to $10^{-3}$ *M* although it was not clear the effect saturated. Xanthopterin (2-amino-4,6-dihydroxypteridine), pterine (2-amino-4-hydroxypteridine) and pterin-6-carboxylic acid were equally effective as folic acid. PABA (paraaminobenzoic acid),

lumazine (2,4-dihydroxypteridine), and folinic acid (5-formyl-tetrahydropteroylglutamic acid) were completely ineffective. The quinazolines (5,8-dideazafolic acid and 5,8-dideazaisofolic acid) which lack the reducible nitrogen atoms in the pteridine ring were partially effective. Since the pteridine ring is apparently the active portion of the folic acid molecule, it was surprising that folate tetraglutamate was completely inactive. Amethopterine (4-amino-10-methyl pteroylglutamic acid) and aminopterine (4-amino pteroylglutamic acid) were less than 20% as effective as folic acid at the highest concentrations tested ($10^{-4}$ *M*). Amethopterine blocked the effect of folic acid when the two were added together, but this inhibition was readily reversible. When the perfusion solution was switched from $10^{-4}$ *M* amethopterine to $10^{-4}$ *M* folic acid, the effect of folic acid was similar to a control not pretreated with amethopterin.

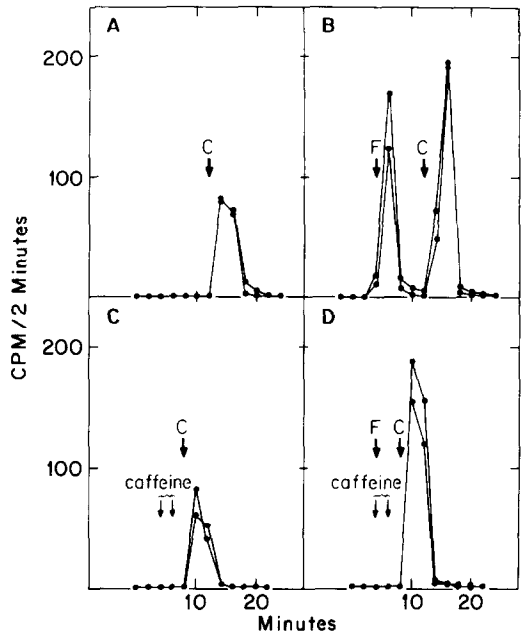


FIG. 7. Effect of pretreatment with folic acid on subsequent responsiveness to cAMP. NC-4 cells were prepared and rates of [$^3$H]cAMP secretion measured as described under Methods. Eight identical sets of cells are represented; experiment illustrated in each of four panels was carried out in duplicate (duplicate symbols). In panels B and D, $10^{-4}$ *M* folic acid was introduced as indicated (F↓) and withdrawn when $5 \times 10^{-9}$ *M* cAMP was introduced as indicated (C↓). In panels C and D, 2 m*M* caffeine was also included in the indicated fractions.

## DISCUSSION

It has been demonstrated that folic acid potentiates cAMP synthesis and secretion in response to cAMP stimuli (Fig. 6); the increase elicited by folic acid plus cAMP was often greater than the sum of the increases elicited by each chemoattractant individually (Fig. 4C). In contrast, folic acid presented singly as stimulus (i.e., under conditions of rapid perfusion in which cAMP cannot serve as a secondary stimulus) elicits only a minimal increase in the rate of [$^3$H]cAMP secretion (Figs. 3A, 4B). Thus, the large response (i.e., elevation in cellular cAMP) elicited by folic acid in cell suspensions is most likely due to the enhancement by folic acid of a response to secreted cAMP (Wurster *et al.*, 1978; Fig. 2).

Often two to three times more cAMP was released when folic acid was added to a high cAMP stimulus (Figs 2, 4D, F, 6). This suggests that the highest doses of cAMP do not maximally activate the adenylate cyclase. It was also noted that folic acid increased the duration of the response to maximal doses of cAMP (Figs. 4D, F, 6A–C). This again indicates that folic acid plus cAMP is interpreted as a larger stimulus since progressively larger cAMP stimuli elicit longer responses (Devreotes and Steck, 1979).

Kawai (1980) has demonstrated that folic acid rapidly increases the binding of [$^3$H]cAMP to both vegetative and aggregation competent cells of *D. discoideum*. Folic acid had similar effects on binding of $10^{-9}$ *M* and $10^{-7}$ *M* [$^3$H]cAMP, suggesting that the number of sites, not the affinity, increased. Such an effect would be consistent with the results presented here. Folic acid, by suddenly increasing in the number of cAMP binding sites available to background cAMP, could initiate cAMP synthesis and secretion (Figs. 2, 3, 4B, E, 5A, F, 7B). Added together with a given cAMP stimulus, it would potentiate cAMP-stimulated cAMP synthesis and secretion by increasing the amount of bound cAMP (Figs. 4C, F, 6). Switching the stimulus from cAMP to folic acid would lead to abrupt termination of the response (Fig. 5); the decrease in the number of occupied cAMP binding sites upon removal of cAMP would be more significant than the increased availability of sites provided by folic acid. However, a second increase in [$^3$H]cAMP secretion is expected if the stimulus is switched to folic acid plus cAMP (Fig. 6B). Although the data presented here support the possibility that folic acid increases cAMP binding to the cell surface, other mechanisms are also possible. In fact, preliminary attempts to repeat a Kawai's (1980) observation have been unsuccessful. Folic acid could also act by inhibiting a vital phosphodiesterase although no effect was found on either the cell membrane phosphodiesterase or total cellular phosphodiesterase (data not shown).

In each of these proposed mechanisms a response to folic acid occurs only in the presence of cAMP. That is, folic acid appears to potentiate a pathway initiated by the binding of cAMP. The slight effects of folic acid

TABLE 1
[$^3$H]cAMP SECRETION ELICITED BY FOLIC ACID AND PTERIDINE ANALOGS

| Stimulus ($10^{-4}$ *M*) | Relative amount of [$^3$H]cAMP secreted |
|---|---|
| Folic acid | 1.0 |
| Xanthopterine | 1.5, 1.1, 1.2, 0.9 |
| Pterine | 0.9, 1.2 |
| Pterine-6-carboxylic acid | 1.2 |
| Paraminobenzoic acid | 0, 0 |
| Lumazine | 0, 0 |
| Folinic acid | 0, 0 |
| 5,8-Dideazafolic acid | 0.5 |
| 5,8-Dideazaisofolic acid | 0.6 |
| Dihydrofolic acid | 0.5 |
| Folate tetraglutamate | 0, 0.05 |
| Amethopterin | 0.05, 0.1, 0.2 |
| Aminopterin | 0.10 |

*Note.* Each value in the table represents the amount of [$^3$H]cAMP secreted during 15 min following application of the stimulus relative to the amount secreted by an identical culture during treatment with folic acid. All responses were measured under conditions of rapid perfusion in the absence of cAMP.

alone on rates of [$^3$H]cAMP secretion under rapid perfusion conditions are most likely due to enhancement of a response to residual low background levels of cAMP. (The greater than additive effect of the mixed stimuli also supports this conclusion.) However, since this slight effect of folic acid could not be completely eliminated (Figs. 3A, C, 4B, 7B), the possibility that it directly elicits a small increase in cAMP synthesis and secretion cannot be ruled out. If this were the case, the effect of $10^{-3}$ *M* folic acid would be equivalent to that of about $10^{-9}$–$10^{-8}$ *M* cAMP. In addition whereas pretreatment with cAMP causes adaptation to both cAMP and folic acid; pretreatment with folic acid does not decrease subsequent responsiveness to cAMP.

Bonner *et al.* (1970) reported that the folic acid potentiated the chemotactic response to low doses of cAMP. Is the effect of folic acid as a chemoattractant merely an enhancement of cAMP-mediated chemotaxis? To examine this possibility the light-scattering response (presumably a reflection of the chemotactic response) elicited by folic acid in suspensions of Ax-3 cells was examined. Advantage was taken of the observation that 2 m*M* caffeine inhibits cAMP secretion but has little effect on the light-scattering response (M. Brenner, personal communication). It was found that folic acid-elicited light-scattering responses were unaffected by caffeine. These preliminary results suggest that while the majority of the effect of folic acid on cAMP production depends on extracellular cAMP, its effect on light scattering is not due to a secondary response to secreted cAMP (experiment carried out with M. Brenner).

This result is puzzling since the studies reported here with structural analogs of folic acid suggest that the site involved in folic acid-directed chemotaxis is similar to that which mediates the effects of folic acid on cAMP synthesis and secretion. For stimulation of cAMP production, the pteridine ring appears to be the active portion of the molecule since pterine and xanthopterine are equally as effective as folic acid while paraaminobenzoyl glutamate is inactive. The 2-position amino group appears essential for activity since lumazine was ineffective. Previous work on chemotaxis led to similar conclusions (Pan and Wurster, 1978). Folinic acid had no effect on cAMP synthesis and secretion suggesting that the 5-position of the pteridine ring is essential for this activity. Although folinic acid is reduced, quinazoline analogs (which cannot be reduced) were partially active (see Table 1). It will be interesting if a similar pattern of activity is found for chemotaxis.

Folic acid apparently serves as a chemoattractant for cells in the early stages of development and also affects cAMP production and secretion in aggregation competent cells, suggesting that the intracellular pathways mediating chemotaxis and cAMP signaling are related. The simplest possibility would be that a cAMP and a folic acid receptor each can stimulate each pathway (Bernstein *et al.*, 1981). However, the results presented here suggest that folic acid and cAMP do not affect cAMP production by identical mechanisms. There is only a minimal activation of cAMP production by folic acid, it does not appear to elicit adaptation as does cAMP, and it appears to potentiate cAMP-mediated cAMP production. Studies are in progress to elucidate these differences and to further understand the relationship of the chemotactic and cAMP-signaling pathways.

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